

AERIX01 receiver modules

Project description and Program of Requirements (PoR) for three plug-in receiver modules and one base-board change

Version 0.1, 2026-09-19. Status: draft for the hardware designer. Normative detail lives in the four Markdown specifications in docs/hardware/modules/ (00-common, 01-subghz, 02-2g4, 03-5g8); where this PDF and those files disagree, the Markdown files win and this PDF must be regenerated (make_por_pdf.py). Priority codes: M = must (acceptance blocker), S = should (deviation needs project-lead sign-off), C = could (nice to have).

Part 1 - Project description

1.1 Why this project exists

AERIX RF is a passive, receive-only drone-detection system. Its primary receiver is a software-defined radio (ANTSDR E200) that captures IQ data around 2.4 GHz and runs energy-morphology, classification and DroneID decoding on a Linux host. That path is deep but narrow: one SDR looks at one band segment at a time, and it costs host CPU and USB bandwidth.

A static analysis of a commercial handheld detector (Tsukorok S3 v4, firmware 5.5.12) showed that a large part of the practical coverage - sub-GHz FSK/LoRa control links (Orlan/Zala/Lancet-class and ELRS-class), 2.4 GHz wideband persistence, and 5.8 GHz video/DJI-class occupancy - is obtained with three cheap narrowband radios doing RSSI sweeps and preamble-triggered packet capture. Those radios (SX127x, SX128x, a 5.8 GHz RSSI receiver) draw about 100-250 mA, need no IQ processing and can run continuously and in parallel.

This project adds that coverage to the AERIX01 sensor as three plug-in modules, one per band, sitting in the spare MCU slots of the existing AERIX01 V1 base board. The goal is not to copy the handheld: the modules report far more than alarms (every packet above a gate, full sweeps, occupancy statistics) so that the always-online server can build per-sensor baselines, whitelists and ODID-labelled training data. The handheld's thresholds become configurable starting points, not the decision.

1.2 System context

Element	Role in this project
AERIX01 V1 base board	Existing. MAIN ESP32-S3 = system controller (also keeps Wi-Fi/BLE scanning). AUX ESP32-S3 slot, two ESP32-C5 slots, USB2517 hub (upstream USB-C to the Linux host), W5500 Ethernet with PoE, MAX-M10S GNSS with TIMEPULSE, mini-PCIe LTE modem.
Module 1 - Sub-GHz	SX1276, 860-1020 MHz. FSK packet capture after preamble (first 4 bytes = "word"), LoRa CAD, 100 kHz RSSI sweeps. Highest value.
Module 2 - 2.4 GHz	SX1280, 2400-2500 MHz (2200-2400 to be verified). 400 kHz RSSI sweeps at 1 sweep/s, DJI-class scoring as candidates, ELRS-2.4 LoRa CAD/packets.
Module 3 - 5.8 GHz	RX5808 / RTC6715, 5645-5945 MHz. Analog-RSSI sweeps (slow, ~6 s), steep-edge candidates for FPV-video/DJI-5.8 class.
MAIN S3	Collects the three module UART streams, time-stamps against GNSS, forwards over Ethernet/PoE (LTE fallback with reduced classes).
Linux host (on-board)	Runs the SDR path; optionally takes bulk traces from the modules over USB (class G).
Server	Baselines, whitelists, differencing between sensors, ODID-labelled learning, final alarm decision. Owns the word tables and thresholds.

Data flow: module radio → module ESP32-S3 (scan loop, gate, candidate rules) → UART 921 600 baud → MAIN S3 (UTC time from GNSS PPS, credits) → Ethernet → server. Bulk traces: module USB → hub → host. Every record carries the PPS count and microseconds since the last PPS edge, so bursts seen by different modules and different sensors can be matched to about 1 µs.

1.3 Scope and deliverables

In scope:

- Three module PCBs (schematic, layout, BOM, assembly drawing, test-pad map) in the existing C5-module outline, interchangeable between the two C5 slots and the AUX slot.
- One base-board change on AERIX01 V1: GNSS TIMEPULSE routed to one pin of each receiver slot (0 Ω options on the module side).
- Firmware per module (ESP-IDF, RadioLib, TinyUSB) implementing the scan plans, candidate rules, the UART/USB data contract and the receive-only lock-out.

- MAIN S3 firmware extension: three UART collectors, credit flow control, CFG/TIME distribution, Ethernet forwarding, LTE class masking.
- Acceptance test procedure and first-article test report per module (Part 2, section T).

Out of scope (do not design for):

- Any transmit, jamming, spoofing, ranging, CAD-then-TX or active interrogation. The modules are receive-only by firmware; TX paths are never initialised and the acceptance test checks the antenna port for emissions.
- Replacing the SDR path: the modules deliver RSSI and demodulated packet heads, no IQ.
- Protocol decoding on the module. Output is stage-1 (RF morphology) and stage-2 (probabilistic) evidence; the server and the SDR path own anything above that.
- FPGA acceleration, a new base board, or changing the MAIN S3 role.

1.4 Decisions already taken

Topic	Decision
Slots	Two ESP32-C5 slots + AUX S3 slot carry the modules; MAIN S3 stays controller. AUX slot conflicts with LTE control (see risks).
Base board	V1 as built, plus the PPS trace to the slots.
Module MCU	ESP32-S3 with PSRAM (WROOM-1-N8R8 or bare chip + 8 MB flash + 8 MB PSRAM). Wi-Fi/BLE disabled in firmware.
Radios	SX1276 (not SX1278), SX1280/SX1281, RX5808 (RTC6715) with SPI mod. No A5133 (the handheld's 5.8 GHz part is likely unpopulated and its driver toggles TX mode).
Data path	UART to MAIN S3 for classes A-F; USB to host for class G bulk.
Uplink	Ethernet/PoE, tens of kB/s per sensor available; LTE fallback profile masks classes C and E.
TX lock-out	Firmware-only (standard radio wiring, TX never initialised, compile-time RX-only flag, image grep test, emission test).
Sub-GHz band	860-1020 MHz.
Diagnostic set	Raw packet heads (32/64 B), full RSSI sweeps, per-channel occupancy; receiver health as low-priority extra.
"Enough" data	All classes A-F at urban worst-case rate simultaneously with 3x link headroom: about 3 kB/s per module vs about 90 kB/s at 921 600 baud.

1.5 Evidence policy (applies to every field name and every alarm)

AERIX RF distinguishes five evidence levels: (1) RF candidate / morphology, (2) probabilistic classification, (3) protocol-specific evidence, (4) validated deterministic decode, (5) operator-provided truth. The modules produce level 1 and 2 only. A module ALARM is a "detector verdict", not a confirmed drone; the server may downgrade it. The firmware and the documentation must not label an RSSI shape or a repeated word as a drone type without the level attached. Whitelisting (5 GHz Wi-Fi channels 149-165 inside the 5.8 GHz band, LoRaWAN/Meshtastic on sub-GHz, Wi-Fi blocks at 2.4 GHz) is done on the server; the module only flags.

1.6 Risks and open items for the designer

#	Item	Owner / how to close
R1	AUX slot vs LTE: the AUX S3 owns the modem UART/control lines. A module in that slot removes LTE unless the modem is driven from MAIN S3 in firmware.	Designer confirms which nets the AUX slot header carries; project lead decides per site (2 or 3 modules).
R2	RX5808 supply: 180-200 mA for the receiver plus ~60 mA ESP32 sits at the 250 mA slot budget; the base ferrite (120 Ω @ 100 MHz) DC rating must be $\geq$ 500 mA.	Designer checks FB10/FB11 rating and 3V3 rail budget; fallback is the AUX slot or an RTC6712/6715 board with lower current.
R3	SX1280 below 2400 MHz (raster B, 2200-2400 MHz) is out of datasheet range; the handheld does it anyway.	Bench measurement on the first article; if RSSI accuracy is unacceptable, raster B is dropped and flagged in HELLO capabilities.
R4	Sub-GHz antenna: most 868/915 whips roll off above 930 MHz; the plan covers 970-1020 MHz.	Antenna selection note; wideband or dedicated antenna for the upper range.
R5	PPS position on the AUX slot header may differ from the C5 slots.	Designer verifies the netlist; 0 Ω options R_PPS_A/B and R_BOOT_A/B on the module.
R6	USB 2.0 hub upstream is shared by everything on the board; class G bulk must not starve the SDR path.	Host throttles class G (module never pushes unrequested bulk).
R7	ESP32-S3 40 MHz clock harmonics (60 x 40 MHz = 2400 MHz) on the 2.4 GHz module.	Layout: ESP32 at the far end from the u.FL, shield can, sensitivity test with the scan loop running.

1.7 Phasing

Phase	Content	Exit criterion
0 Design review	Designer reviews the four Markdown specs + this PoR, returns questions and the AUX/RX5808/PPS netlist answers (R1, R2, R5).	Open items R1, R2, R5 closed in writing.
1 Schematic + layout	Three modules, base-board PPS ECO. Test pads per spec.	Design review against Part 2 sections M, E, I.
2 First article	Two units per module type, hand-assembled or small batch.	Fit in all three slots; enumerates; boots; HELLO on UART.
3 Firmware bring-up	Scan plans, data classes, PPS timestamping, RX-only build; MAIN S3 collector.	Common tests T-01...T-06 pass; module tests pass on the bench.
4 Field pilot	One sensor with all three modules online for ≥ 2 weeks in an urban site; server collects classes A-F.	Baseline/occupancy data usable; false-alarm rate report; go/no-go for batch.

Part 2 - Program of Requirements

Each requirement has an ID (section letter + number), a verification method and a priority. Numeric values are the defaults from the Markdown specifications; "CFG" means the value is a runtime configuration item owned by the MAIN S3/server.

G - General

ID	Requirement	Verification	Prio
G-01	The modules are receive-only. No transmit, ranging, CAD-then-TX, beacon or interrogation function of any radio may be enabled in hardware or firmware.	Design review; T-06 emission test; firmware image grep for TX entry points.	M
G-02	Three module types: Sub-GHz (SX1276), 2.4 GHz (SX1280/1281), 5.8 GHz (RX5808/RTC6715). One PCB design per type.	BOM review.	M
G-03	A module of any type fits and functions in either ESP32-C5 slot and in the AUX ESP32-S3 slot of AERIX01 V1 without rework other than 0 Ω option resistors.	T-01 slot fit.	M
G-04	All module output is tagged with evidence level 1 or 2; no field or document may present a module alarm as a confirmed drone.	Documentation and protocol review.	M
G-05	Default thresholds and scan plans reproduce the reference handheld (firmware 5.5.12 extraction brief) where stated, but every one of them is a CFG item.	Firmware review; CFG round-trip test.	M
G-06	Deliverables per module: schematic, layout, BOM, assembly drawing, test-pad map, first-article test report against section T.	Document check at hand-over.	M

M - Mechanical

ID	Requirement	Verification	Prio
M-01	Outline, mounting and header positions identical to the existing ESP32-C5 module drawing (designer holds the master outline).	Overlay of the two outlines; T-01.	M
M-02	Headers: two 1x12 2.54 mm and two 1x4 2.54 mm, base side PPTC121LFBN-RC and 61300411821, same positions as the C5 module.	Layout check.	M
M-03	One u.FL (IPEX MHF1) at the same corner as the existing modules; 50 Ω ground-stitched microstrip, $\leq$ 15 mm from the radio pin to the connector.	Layout check; TDR or return-loss spot check on first article.	M
M-04	Shield can over the radio section (radio + match + filter) on the Sub-GHz and 2.4 GHz modules; ground fence around the RSSI/ADC trace on the 5.8 GHz module.	Layout check.	M
M-05	Height envelope as the existing module (shield can + u.FL). No components on the base-board side of the module.	Mechanical check in all three slots.	M
M-06	Silkscreen: module type, revision, slot mark "C5/AUX".	Visual.	S
M-07	Test pads: SCK/MISO/MOSI/NSS and DIO0 (SX modules), RSSI analog (5.8 GHz), GPIO0, EN, UART TX/RX, PPS, 3V3, GND.	Layout check.	S

E - Electrical and power

ID	Requirement	Verification	Prio
E-01	The module uses only these base-side nets: 3V3 (left 1x12 pin 12), EN (left pin 10), GND (left pins 1 and 6, USB 1x4 pins 1 and 4), UART0 TX/RX (right 1x12 pins 10/9), BOOT (right pin 3 via 0 Ω), USB D-/D+ (USB 1x4 pins 2/3), PPS (right pin 6 via 0 Ω). Every other header pin is no-connect on the module.	Netlist review against the base board for all three slots.	M
E-02	Where BOOT or PPS positions differ between the C5 and AUX slots, 0 Ω option resistors (R_BOOT_A/B, R_PPS_A/B) select the position so one PCB serves all slots.	Netlist review.	M
E-03	Supply: single 3V3 through the base ferrite. No back-feed of 5 V; no draw from USB VBUS; VBUS sense only via a divider for connect detection.	Schematic review; T-02.	M
E-04	Current: $\leq$ 250 mA average while scanning, $\leq$ 500 mA for 10 ms peaks (including boot inrush).	T-02 current measurement.	M
E-05	Decoupling: 10 μ F + 100 nF at the module supply input; radio on a separate LDO or filtered branch with $\geq$ 40 dB rejection at 100 kHz-10 MHz (3.3 V to 3.0 V LDO acceptable where the radio allows).	Schematic review; T-03 sensitivity with the scan loop running.	M

ID	Requirement	Verification	Prio
E-06	EN: 10 kΩ to 3V3 and 1 μF to GND on the module; base RST header shorts EN to GND.	Schematic review.	M
E-07	Strapping: GPIO0 on a module push button and test pad; GPIO46 low; GPIO45 low (3.3 V flash).	Schematic review; boot test.	M
E-08	USB D-/D+ routed as a 90 Ω differential pair with an ESD diode array at the header.	Layout review.	M
E-09	ESP32-S3 with >= 8 MB flash (two OTA slots) and >= 8 MB PSRAM (sweep/occupancy buffers). Wi-Fi/BLE unused; a bare chip or a WROOM with the antenna unused is acceptable; the module antenna is never shared with the ESP32.	BOM review.	M
E-10	Radio reference: Sub-GHz 32 MHz TCXO ±2 ppm (crystal ±10 ppm acceptable); 2.4 GHz 52 MHz TCXO ±2 ppm (crystal ±10 ppm acceptable); TCXO preferred on both.	BOM review.	S
E-11	Layout of the 2.4 GHz module places the ESP32-S3 and its 40 MHz crystal at the far end from the u.FL.	Layout review.	S

I - Interfaces (UART, USB, PPS)

ID	Requirement	Verification	Prio
I-01	UART0 to the MAIN S3: 3.3 V, 921 600 baud, 8N1, no RTS/CTS; 115 200 fallback at boot, speed set by CFG.baud.	T-04 UART soak test.	M
I-02	Framing: COBS with 0x00 delimiter, CRC-16/CCITT (poly 0x1021, init 0xFFFF) over the decoded payload, maximum frame 512 bytes.	Protocol test vectors.	M
I-03	Flow control is credit based: the module sends at most the number of frames granted in ACK.credits and drops the lowest-priority classes first when credits are exhausted.	T-04 with artificial credit starvation; HEALTH counters.	M
I-04	Frame header: u8 type, u8 seq (wrap 255), u32 t_us (μs since last PPS edge, or since boot if no PPS), u16 pps_count; fixed little-endian bodies, no strings except LOG.	Protocol test vectors.	M
I-05	Message types: 0x01 HELLO, 0x02 CFG, 0x03 TIME, 0x04 ACK, 0x10 ALARM, 0x11 PACKET, 0x12 SWEEP (chunked), 0x13 OCCUPANCY, 0x14 CANDIDATE, 0x1F HEALTH, 0x20 CMD, 0x7F LOG.	Protocol review.	M
I-06	Units: frequency Hz (u32), RSSI/SNR 0.25 dB steps (i16) in records, i8 dBm in sweep vectors, bitrate bps (u32), bandwidth Hz (u32), duration μs (u32).	Protocol review.	M
I-07	USB: ESP32-S3 native USB (TinyUSB), self-powered; one CDC-ACM interface carrying the same framed protocol and one vendor bulk endpoint for class G. Enumeration must not depend on VBUS.	T-01 enumeration; host bulk test.	M
I-08	The module runs fully on UART alone; USB is optional at runtime. Class G is sent only on host request and is throttled by the host.	Run with the USB pins disconnected.	M
I-09	PPS input on GPIO4 via 0 Ω from right 1x12 pin 6; captured by ISR or GPTimer capture; t_us resets on each edge. Without PPS the module syncs to TIME messages (±1 ms) and flags "no PPS" in HEALTH.	T-05 two-module timestamp test.	M

D - Data contract ("enough")

The link must carry all classes A-F at their nominal urban worst-case rate simultaneously, with 3x headroom. This is about 3 kB/s per module against about 90 kB/s available; three modules use three separate UARTs on the MAIN S3.

Class	Content	Nominal rate	Bytes/s	Prio (1 = dropped last)
A ALARM	type code, freq, bandwidth/bitrate, RSSI, SNR, score, fingerprint (word or hash), count, first/last time	<= 2/s	200	1
B PACKET	every packet above the gate: freq, bitrate/SF/BW, RSSI, SNR, frequency error, preamble length, first 32 bytes (64 on request), CRC-present flag	20/s FSK, 5/s LoRa	2 000	2
C SWEEP	full RSSI vector, i8 dBm: sub-GHz 1 600 pts / 5 s; 2.4 GHz 250 pts / 1 s; 5.8 GHz 175 pts / 6 s	per module	320 / 250 / 30	4
D OCCUPANCY	per channel: busy fraction u8, RSSI p50/p95 i8, burst count u8; delta once per 60 s	1/min	<= 110	3

Class	Content	Nominal rate	Bytes/s	Prio (1 = dropped last)
E CANDIDATE	stage-1 events without alarm: freq, width, duration, score, why-rejected code	<= 5/s	250	3
F HEALTH	noise floor per band, gain state, radio temperature, PLL lock, drop and credit-starvation counters, PPS present, fw hash	1 / 10 s	10	1
G TRACE	USB only: time-domain RSSI traces (e.g. 8 192 samples), long packet bodies, high-rate sweeps	on request	<= 200 kB/s	USB

ID	Requirement	Verification	Prio
D-01	Every module implements classes A, C, D, E, F; the Sub-GHz and 2.4 GHz modules also implement B; the 5.8 GHz module has no B.	Protocol test per module.	M
D-02	Class B is emitted for every packet above the gate regardless of detector outcome (the near-alarm stream).	Bench transmitter test (T-R1-3).	M
D-03	Class C sweeps carry the exact start and stop timestamp of the sweep.	Protocol review.	M
D-04	Class enable mask, packet head length (32/64) and cadences are CFG items; the LTE fallback profile switches classes C and E off.	CFG test.	M
D-05	Raw RSSI and the applied calibration offset are reported separately; no fixed correction (e.g. the handheld's +15 dB) is baked into values.	Protocol review.	M

F - Firmware, common

ID	Requirement	Verification	Prio
F-01	ESP-IDF 5.x, RadioLib for the SX radios, TinyUSB; Arduino component optional.	Build review.	S
F-02	Compile-time CONFIG_AERIX_RX_ONLY=y removes TX code paths; radio TX pins and PA control stay in reset state; a unit test greps the linked image for TX entry points.	CI test.	M
F-03	HELLO within 2 s of EN release (module type, hw rev, fw version/hash, capability bitmap, PPS present, calibration). Scanning starts after CFG, or after 5 s from NVS defaults if the MAIN S3 is silent.	Boot test.	M
F-04	Watchdog on the scan loop; HEALTH every 10 s; no dynamic allocation inside the scan loop.	Code review; 1 h soak (T-04).	M
F-05	NVS holds band plan, gates, cadences, module serial, per-band RSSI calibration offsets.	CFG persistence test.	M
F-06	OTA through the MAIN S3 (CMD reboot-to-bootloader + UART loader) and through USB.	OTA test both paths.	S
F-07	Wi-Fi and BLE on the module ESP32-S3 are disabled.	Code review; current measurement.	M

R1 - Module 1, Sub-GHz SX1276 (860-1020 MHz)

ID	Requirement	Verification	Prio
R1-01	Radio SX1276 (862-1020 MHz variant; not SX1278, not SX1262). Antenna to RFI_HF via the datasheet HF match; RFO_HF/PA_BOOST unconnected; no RF switch.	BOM/schematic review.	M
R1-02	Front-end filter passing 860-1020 MHz (e.g. 5th-order high-pass ~800 MHz + low-pass ~1.1 GHz) to limit GSM-900 downlink overload. The 925-960 MHz band stays inside the sweep plan.	T-R1-2 overload test.	M
R1-03	Optional LNA footprint (1 dB NF class) with 0 Ω bypass, not fitted in v1.	Layout review.	C
R1-04	SPI <= 10 MHz plus RESET, DIO0, DIO1, DIO2 on ISR-capable GPIOs; DIO5 optional.	Schematic review.	M
R1-05	Antenna note: 868/915 whips roll off above 930 MHz; a wideband or dedicated antenna is required for 970-1020 MHz coverage.	Antenna selection document.	S
R1-06	Default band plan 860-885, 895-928, 970-1020 MHz at 100 kHz (1 080 channels); accepts up to 8 ranges inside 860-1020 MHz.	CFG test.	M
R1-07	FSK reception at 57.6, 80.0, 15.235 kbps (defaults) plus 38.4, 76.19, 38.15 kbps; RX BW 200 kHz, AFC on, preamble detector on, sync-word detection off so the first 4 bytes after the preamble are captured as the "word".	T-R1-3 word capture.	M

ID	Requirement	Verification	Prio
R1-08	LoRa CAD at 500 kHz BW SF6-9 CR 4/7 sync 0x12; also 125/250 kHz SF7-12 sync 0x12 and 0x34 for LoRaWAN/Meshtastic inventory (identify only).	T-R1-4.	M
R1-09	Scheduler: RSSI pre-pass ($\geq 140 \mu\text{s}$ dwell, $\sim 0.2 \text{ s}$), FSK dwell 50 ms per bitrate only on channels above the gate, LoRa CAD on the same channels, full class C sweep every 5 s. Full cycle $\leq 60 \text{ s}$ quiet, $\leq 3 \text{ min}$ urban.	T-R1-5 cycle time.	M
R1-10	Detection: CFG-loadable static word table with normalisation (top bits 00/11 $\rightarrow$ mask 0x7FFFFFFF, else strip leading alternating bits); repetition detector (same word, same channel, ≥ 2 in 20 min $\rightarrow$ CANDIDATE); ALARM only for table or repetition hits with RSSI $\geq$ gate + 6 dB.	Firmware test with recorded words.	M
R1-11	Sensitivity at 868.3 MHz, 38.4 kbps GFSK, 1 % PER: $\leq -104 \text{ dBm}$ at the u.FL.	T-R1-1.	M
R1-12	Overload: -20 dBm CW at 942 MHz degrades 868.3 MHz sensitivity by $\leq 6 \text{ dB}$.	T-R1-2.	M

R2 - Module 2, 2.4 GHz SX1280 (2400-2500 MHz)

ID	Requirement	Verification	Prio
R2-01	Radio SX1280 or SX1281; RX only; optional 2400-2500 MHz band-pass ($\leq 1.5 \text{ dB IL}$); no RF switch, or the reference SPDT tied permanently to RX.	BOM/schematic review.	M
R2-02	Optional 2.4 GHz LNA footprint with 0Ω bypass, not fitted in v1.	Layout review.	C
R2-03	SPI $\leq 18 \text{ MHz}$ plus RESET, BUSY, DIO1 (DIO2 optional).	Schematic review.	M
R2-04	Sweep raster A: 2400.0-2499.6 MHz, 400 kHz, 250 channels, $40 \mu\text{s}$ settle, RSSI in FSK mode; one sweep per second for class C, eight back-to-back every 5 s for scoring.	T-R2-1 sweep timing.	M
R2-05	Sweep raster B: 2200.0-2399.6 MHz, 400 kHz, $60 \mu\text{s}$ settle, every 5 s when enabled. Feasibility below 2400 MHz must be measured; if it fails the capability bit is cleared in HELLO.	T-R2-5.	S
R2-06	Fine raster: $256 \times 390.625 \text{ kHz}$ from 2400 MHz selectable instead of raster A.	CFG test.	S
R2-07	LoRa at 2.4 GHz: BW 812.5 kHz, SF5-8, CR 4/6 (SF5) or 4/8, sync 0x14 and 0x12, preamble 12 (ELRS 2.4); 400 kHz steps over a CFG range $\leq 100 \text{ MHz}$; per-packet RSSI, SNR, frequency error, first 32 bytes.	T-R2-4.	M
R2-08	Track mode on CMD: $f \pm 10 \text{ MHz}$ at 400 kHz, $80 \mu\text{s}$ settle, two passes.	CMD test.	S
R2-09	Detection: 16-channel window rule (≥ 8 of 16 above gate, broadband guard < 125 of 250), spectral score A (median + 5 dB, width 26-52 bins, persistence ≥ 3 of 8) and score B (width 18-60 bins, 85 % fill, 5 dB centre-edge margin) as candidate producers; ALARM only after ≥ 2 consecutive scoring passes above the CFG threshold. Noise histogram in HEALTH only.	Firmware test with recorded sweeps.	M
R2-10	Sweep timing: 250 channels in $\leq 15 \text{ ms}$ including settle; RSSI repeatability $\pm 1.5 \text{ dB}$ at -60 dBm.	T-R2-1.	M
R2-11	RSSI linearity -100 to -30 dBm within $\pm 3 \text{ dB}$ at 2440 MHz.	T-R2-2.	M
R2-12	PSRAM sized for 8 sweeps $\times$ 256 bins plus the 250-bin history.	Memory budget review.	M

R3 - Module 3, 5.8 GHz RX5808 / RTC6715 (5645-5945 MHz)

ID	Requirement	Verification	Prio
R3-01	Receiver RX5808 (RTC6715) in SPI mode (fixed-channel resistor removed, CH1/CH2/CH3 as SPI CLK/DATA/LE); shielded variant with u.FL preferred, otherwise a short 50Ω trace to a u.FL. Alternatives: any RTC6715/RTC6712 board with SPI synthesizer and RSSI pin.	BOM review.	M
R3-02	Tuning range 5645-5945 MHz in $\sim 1 \text{ MHz}$ steps (~ 300 channels); default plan 5725-5899 MHz, 175 channels.	T-R3-1.	M
R3-03	RSSI path: analog RSSI $\rightarrow$ RC $1 \text{ k}\Omega / 10 \text{ nF}$ $\rightarrow$ ESP32-S3 ADC1; test pad; calibrated attenuation or precision reference; ADC noise $\leq 1 \text{ LSB rms}$ at 12 bits after the filter.	T-R3-2.	M
R3-04	Video output not routed; terminated per module datasheet.	Schematic review.	M

ID	Requirement	Verification	Prio
R3-05	Supply: RX5808 180-200 mA + ESP32-S3 ~60 mA. The designer confirms the base 3V3 rail and ferrite (DC rating >= 500 mA) support 250 mA; otherwise the AUX slot or a lower-current RTC67xx board is used. There is no 5 V on the slot.	T-02; ferrite datasheet check.	M
R3-06	Per channel: write synthesizer, wait for lock (measured, default 30 ms), RSSI = mean of 16 ADC samples over 2 ms. Sweep ~5.6 s default, ~10 s extended; class C cadence one sweep per 6 s.	T-R3-3.	M
R3-07	Adaptive mode: after a full sweep, re-visit up to 16 channels above the gate every 1 s; full sweep every 6 s. Gate default floor + 10 dB where floor = quietest channel of the first sweep.	Firmware test.	S
R3-08	Detection: steep-edge rules (thresholds -65 to -99 dBm in 5 dB steps, gap tolerance 5 channels, width 16-69 MHz, edges stable within 5 MHz, two consecutive sweeps) and the 16-channel window rule as candidate producers; ALARM only after >= 3 sweeps (~18 s) and not on a whitelisted channel; no time-domain classification on this module.	Firmware test with recorded sweeps; T-R3-5.	M
R3-09	Class G over USB: continuous RSSI ADC stream on one channel up to 10 kS/s.	Host bulk test.	C
R3-10	RSSI curve monotonic -95 to -40 dBm within ±3 dB after two-point calibration stored in NVS and reported in HELLO.	T-R3-2.	M
R3-11	LO leakage at the antenna port below -57 dBm.	T-R3-6 / T-06.	M

B - Base-board change (AERIX01 V1 ECO)

ID	Requirement	Verification	Prio
B-01	Route GNSS TIMEPULSE (MAX-M10S pin 4, series 33 Ω R86, today only on the MAIN S3 P3 header) to one currently unconnected header pin of each of the three receiver slots. Proposed: right 1x12 pin 6 (C5 "GPIO15" position); verify the AUX equivalent is free.	Netlist review.	M
B-02	Series 33 Ω per branch; 3.3 V CMOS, 100 ms high pulse once per second. If any branch exceeds ~15 cm, add one 74LVC1G17 buffer near the GNSS.	Layout review; T-05.	M
B-03	No other base-board change; V1 stays as built.	ECO review.	M
B-04	MAIN S3 firmware: three UART collectors at 921 600 baud, credits, CFG/TIME distribution (TIME once per second after PPS), UTC conversion of pps_count/t_us, Ethernet forwarding of >= 3 x 3 kB/s, LTE class masking.	End-to-end test with three modules.	M

T - Acceptance tests (first article, before sign-off)

Common tests, each module type:

ID	Test	Pass criterion
T-01	Slot fit and bring-up	Fits both C5 slots and the AUX slot; RST/BOOT headers work; enumerates on the USB2517 hub; HELLO within 2 s.
T-02	3V3 current during scanning	<= 250 mA average, <= 500 mA peak (10 ms), including boot.
T-03	Sensitivity with the scan loop running	Within 3 dB of the radio datasheet at the u.FL (proves supply filtering and layout).
T-04	UART soak	1 h at 921 600 baud with classes A-F at nominal rate: zero CRC errors, zero credit starvation.
T-05	PPS timestamping	Two modules on one base board timestamp a common test burst within 2 μs.
T-06	Spurious emission	No measurable emission at the antenna port in any band (spectrum analyser, RBW 10 kHz); confirms receive-only in hardware.

Module-specific tests:

ID	Test	Pass criterion
T-R1-1	Sub-GHz sensitivity	868.3 MHz, 38.4 kbps GFSK, 1 % PER: <= -104 dBm.
T-R1-2	Sub-GHz overload	-20 dBm CW at 942 MHz: sensitivity loss <= 6 dB.
T-R1-3	Word capture	Bench generator sending a known 32-bit word at 57.6 kbps: same word in >= 99 % of class B records; frequency error within ±2 kHz.
T-R1-4	LoRa CAD	SF7 / 500 kHz packets at -110 dBm detected on >= 90 % of passes.
T-R1-5	Cycle time	<= 60 s with no activity; <= 180 s with 50 active channels.

ID	Test	Pass criterion
T-R2-1	2.4 GHz sweep timing	250 channels in ≤ 15 ms; RSSI repeatability ± 1.5 dB at -60 dBm CW.
T-R2-2	2.4 GHz RSSI linearity	-100 to -30 dBm within ± 3 dB at 2440 MHz.
T-R2-3	Wi-Fi block behaviour	20 MHz Wi-Fi at -50 dBm appears as a ~51-bin block and is reported as CANDIDATE, never ALARM, at defaults.
T-R2-4	ELRS profile	SF5 / 812.5 kHz packets at -100 dBm received ≥ 90 %; frequency error within ± 5 kHz.
T-R2-5	Raster B feasibility	RSSI accuracy at 2200-2400 MHz measured and recorded; capability bit set accordingly.
T-R2-6	No emission	T-06 including no CAD-then-TX and no ranging.
T-R3-1	5.8 GHz tuning	Every channel of the extended plan locks; lock time ≤ 35 ms.
T-R3-2	5.8 GHz RSSI curve	Monotonic -95 to -40 dBm at 5800 MHz within ± 3 dB after calibration.
T-R3-3	5.8 GHz sweep time	≤ 6.5 s default plan, ≤ 11 s extended plan.
T-R3-4	5.8 GHz current	≤ 250 mA average; inrush ≤ 500 mA / 10 ms.
T-R3-5	5 GHz Wi-Fi behaviour	20 MHz Wi-Fi channel at -50 dBm reported as CANDIDATE (steep-edge rule), never ALARM at defaults.
T-R3-6	LO leakage	Below -57 dBm at the antenna port.

Appendix - Reference documents

- docs/hardware/modules/00-common-slot-interface.md - slot interface, UART/USB contract, data classes, base-board ECO, common firmware and tests.
- docs/hardware/modules/01-subghz-sx1276-module.md, 02-2g4-sx1280-module.md, 03-5g8-rx5808-module.md - per-module hardware, scan plans, detection rules, tests.
- research/briefs/tsukorok-firmware-5.5.12-rf-extraction.md - origin of the default thresholds, word tables and scan plans (all values with firmware addresses).
- research/briefs/tsukorok/Tsukorok_5.5.12_detection_methods_summary.pdf and Tsukorok_replication_hardware.pdf - detection-method summary and hardware options.
- docs/design/server-baseline-whitelist-plan.md - what the server does with classes B-E (baselines, whitelists, ODID-labelled learning).
- PM_AERIX01_V1.PDF - base-board schematic (sheets Dual ESP32-C5, Dual ESP32S3, USB HUB, GNSS).